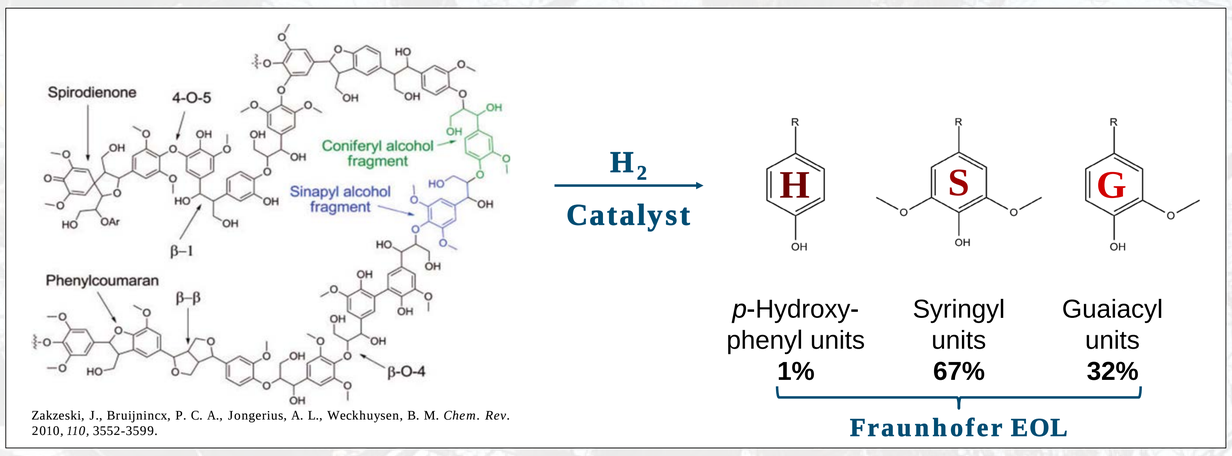
Ethanol Organosolv Lignin Depolymerisation: Development of the Product Treatment and Analysis

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Introduction

In the bio-based economy, lignin is the most potential resource for aromatic chemicals and fuels, and catalysis is regarded to be the key technology for its valorisation [1]. In addition, hydrogenolysis is considered to be the most promising method to produce phenols from lignin [2]. In this work, various noble metal catalysts were screened in lignin depolymerisation with hydrogen. Ethanol was used as a solvent and a capping agent [3]. Ethanol organosolv lignin (EOL) was selected as a feedstock because it is a sulphur-free, low-molecular-weight product of high purity [4]. An essential part of the study is the development of the product treatment and analysis.

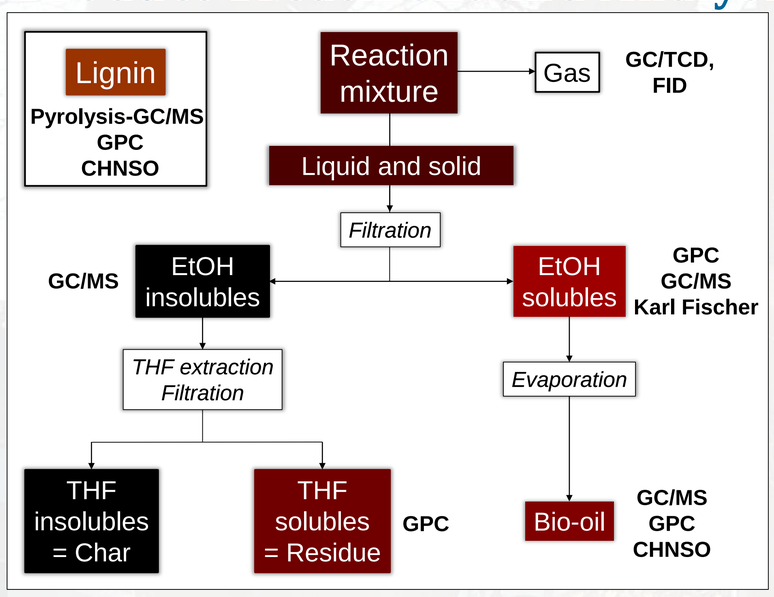


Zakzeski, J., Bruijnincx, P. C. A., Jongerius, A. L., Weckhuysen, B. M. *Chem. Rev.* 2010, *110*, 3552-3599.

Experimental

- 100 ml batch reactor equipped with a spinning basket stirrer (1000 rpm)
- 1 g hardwood (beech) ethanol organosolv lignin (EOL) from the Fraunhofer process in 50 ml EtOH
- 200–400 °C, 10–50 bar H_2 (cold pressure), reaction time up to 3 h
- 100 mg noble metal catalysts with the particle size of 0.30–0.42 mm
  - Commercial catalysts: 3% Pd/C (Degussa) and 5% Pd/Al_2O_3 (Strem Chemicals)
  - ZrO_2 (MEL Chemicals ECO100E) catalysts prepared by impregnation by ourselves

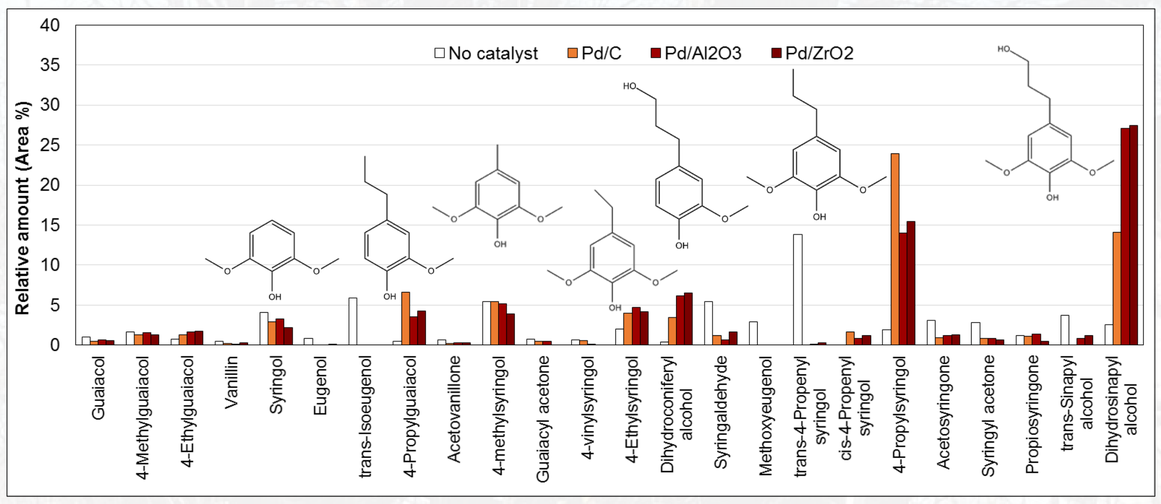
Product Treatment and Analysis



Analysis of Monoaromatic Compounds

- Quantitative analysis challenging
  - Large number of different compounds
  - Partly evaporated in the product treatment
  - Partly poor volatility in GC/MS
- Amounts estimated according to the GC/MS and GPC results

Results – Monoaromatic Compound Distribution



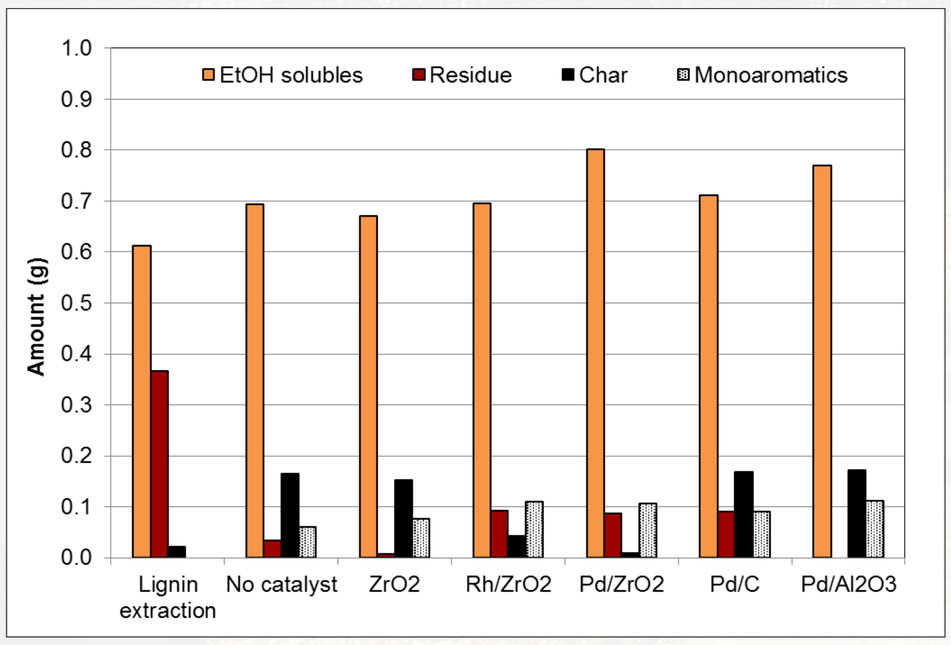
Effect of the Catalysts

- Amount of monoaromatic compounds increased
- Clearly more hydrogenation and hydrodeoxygenation products
- Rh/ZrO_2: higher amount of products with a shorter side chain compared to Pd/ZrO_2
- Pd/oxide catalysts: less -OH group in propyl side chain compared to Pd/carbon

Results – Main Product Fractions

Catalysts increase the amounts of ethanol solubles and monoaromatic compounds

ZrO_2 supported noble metal catalysts decrease the amount of char



Conclusions

- Hydrotreatment of EOL on the noble metal catalysts has a large potential in lignin valorisation
- Catalysts have a remarkable effect on the product distributions
- Clear differences between the catalyst metals and supports
- Pd/ZrO_2 catalyst particularly very good, strong metal support interaction assumed
- More catalyst and process development still required
- Product treatment and analysis development still needed

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